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Physiologically-Based Pharmacokinetic Model-Informed Labeling for Cariprazine Drug Interactions With CYP3A Inhibitors

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Correspondence: Hao Xiong (hao.xiong@abbvie.com; clinpharm.publications@abbvie.com)**Received:** 10 November 2025 | **Revised:** 6 March 2026 | **Accepted:** 11 March 2026**Keywords:** cariprazine | CYP3A4 | drug–drug interactions | PBPK modeling

ABSTRACT

Cariprazine is a potent D_3 -preferring dopamine D_3/D_2 receptor partial agonist, a serotonin 5-HT_{1A} receptor partial agonist, and a serotonin 5-HT_{2B} receptor antagonist approved for the treatment of a variety of psychiatric disorders. A clinical study examining short-term (4 days) drug–drug interactions (DDIs) between cariprazine and ketoconazole, a strong CYP3A4 inhibitor, guided cariprazine dosing adjustment recommendations in concomitant use with CYP3A4 inhibitors in the original US FDA marketing approval. However, didesmethyl-cariprazine (DDCAR), a major active metabolite of cariprazine, takes 4–8 weeks to reach steady-state plasma concentration. Therefore, longer term clinical DDI studies would be needed to fully understand cariprazine DDIs but are greatly challenging. Regulatory agencies are increasingly encouraging the use of physiologically based pharmacokinetic (PBPK) modeling to evaluate DDIs. Here, we developed PBPK models of cariprazine, DCAR, and DDCAR that adequately described their plasma exposures across multiple Phase 1 or 2 clinical studies with cariprazine treatment alone or in combination with CYP3A inhibitors ketoconazole or erythromycin. The validated models predicted up to 6.0-fold, 2.9-fold, and 1.1-fold increases in total cariprazine (cariprazine + DCAR + DDCAR) exposure at steady state upon prolonged coadministration of strong, moderate, and weak CYP3A4 inhibitors, respectively. The PBPK models allowed for more optimal cariprazine dose adjustments with short-term and long-term concomitant use of strong and moderate CYP3A4 inhibitors. Model predictions led to an update in US prescribing information in November 2024 to inform on optimal cariprazine dose adjustment with concomitant use of CYP3A inhibitors. Updated recommendations had the objective of maintaining treatment efficacy while minimizing drug adverse effect risk.

1 | Introduction

Cariprazine is a potent D_3 -preferring dopamine D_3/D_2 receptor partial agonist, a serotonin 5-HT_{1A} receptor partial agonist, and a serotonin 5-HT_{2B} receptor antagonist. The drug is orally administered and approved in the US for the treatment of schizophrenia (1.5–6 mg/day), manic or mixed episodes associated with bipolar I disorder (3–6 mg/day), depressive episodes associated with bipolar I disorder (1.5 or 3 mg/day), and major depressive disorder (adjunctive therapy, 1.5 or 3 mg/day)

in adult patients [1]. The drug is also approved for pediatric use in patients with schizophrenia (13–17 years, 1.5–4.5 mg/day) or manic or mixed episodes associated with bipolar I disorder (10–17 years, 3 or 4.5 mg/day) [1].

Cariprazine is metabolized by cytochrome P450 (CYP) 3A4, and to some extent by CYP2D6, into desmethyl-cariprazine (DCAR) [1–3]. CYP3A4 and CYP2D6 further metabolize DCAR into didesmethyl-cariprazine (DDCAR), which in-turn is metabolized by CYP3A4 into a hydroxylated metabolite [2, 4].

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Study Highlights

- What is the current knowledge on the topic?
 - A clinical study examining short-term (4 days) cariprazine/ketoconazole drug–drug interactions (DDI) guided cariprazine dosing adjustment recommendations in the original US NDA approval in 2015.
- What question did this study address?
 - Due to the limited data on cariprazine interactions with CYP3A inhibitors, optimal cariprazine dosage adjustments at steady-state could not be recommended, especially in the setting of long-term coadministration of cariprazine with CYP3A4 inhibitors.
- What does this study add to our knowledge?
 - Physiologically-based pharmacokinetic (PBPK) models of cariprazine, DCAR, and DDCAR were developed and validated, allowing for subsequent prediction of CYP3A inhibitor/inducer effects on cariprazine exposures for scenarios that were not previously studied in humans.
- How might this change drug discovery, development, and/or therapeutics?
 - PBPK model-based DDI predictions informed cariprazine dosing adjustment recommendations, leading to a US prescribing information update (November 2024) to maintain optimal exposure for efficacy and safety. This analysis highlights how PBPK modeling expands DDI assessment to difficult-to-study scenarios.

Both DCAR and DDCAR have similar binding properties and pharmacological activity as the parent drug [1, 5]. Because cariprazine is largely metabolized by CYP3A4, it was expected to be susceptible to drug–drug interactions (DDIs) with CYP3A4 inhibitors. In a clinical DDI study, coadministration of ketoconazole (400 mg/day for 4 days), a strong CYP3A4 inhibitor, with cariprazine (0.5 mg/day) increased total cariprazine (sum of cariprazine, DCAR, and DDCAR) peak concentration ($C_{\max}$) and area under the plasma concentration–time curve over a 24-h dosing interval (AUC_{0-24h}) by approximately 100% [1]. As a result, initial cariprazine recommendations called for a 50% dose decrease in patients also taking a strong CYP3A4 inhibitor [6]. Subsequently, an investigation of cariprazine (1.5 mg once per day [QD]) and erythromycin (moderate CYP3A4 inhibitor; 500 mg twice daily) coadministration over 21 days revealed a 40%–50% increase in total cariprazine exposure ($C_{\max}$, AUC_{0-24h}) over cariprazine administered alone [3].

Cariprazine, DCAR, and DDCAR pharmacokinetics have been well-characterized [7, 8] and are dose-proportional (linear pharmacokinetics) at steady state over a dosing range of 1.5–12.5 mg/day [2]. $C_{\max}$ of cariprazine occurred approximately 3–6 h after administration of a single cariprazine dose under fasting conditions; a high-fat meal had no significant effect on cariprazine exposures ($C_{\max}$, AUC) [1]. The half-lives based on time to reach steady state are 2–4 days for cariprazine, about

1–2 days for DCAR, and approximately 1–3 weeks for DDCAR [1]. Mean concentrations of DCAR and DDCAR are approximately 30% and 400%, respectively, of cariprazine concentrations at steady state [1].

Some strong or moderate CYP3A4 inhibitors such as certain antifungal and antiviral therapies may require treatment over long periods of time. For example, ketoconazole, a strong CYP3A inhibitor, is a commonly used antifungal therapy with a usual treatment duration of 6 months for systemic infection [9]. However, clinical DDI studies with cariprazine have been relatively short and conducting ones of a longer, more optimal duration is greatly challenging due to decreased participant retention and compliance, increased risk of adverse events, and resource limitations. Here, we developed physiologically-based pharmacokinetic (PBPK) interconnected models for cariprazine, DCAR, and DDCAR pharmacokinetics. The models were then used to predict DDIs with CYP3A4 inhibitors and evaluate dose adjustments for cariprazine with the continuous use of CYP3A inhibitors. No clinical DDI studies of cariprazine with CYP3A4 inducers have been conducted. Thus, cariprazine's DDIs with strong and moderate CYP3A4 inducers were also simulated with the PBPK models. Study results informed a US prescribing information update, including dose adjustment recommendations.

2 | Methods

This analysis utilized existing in vivo pharmacokinetic data. The Independent Ethics Committee or Institutional Review Board at each study site approved the study protocol, informed consent forms, and recruitment materials before patient enrollment. The studies were conducted in accordance with the International Conference for Harmonization guidelines, applicable regulations, and the Declaration of Helsinki. All patients provided written informed consent before screening.

2.1 | Physiologically-Based Pharmacokinetic Model Development

Simcyp (V20.0.157.0, Certara, Sheffield, UK) was used to develop and verify the cariprazine, DCAR, and DDCAR PBPK models. Models for cariprazine, DCAR, and DDCAR were developed independently by incorporating physicochemical properties of the three active moieties (molecular weight, hydrophobicity [logP], and pKa), plasma protein binding properties, blood–plasma partitioning, and cellular permeabilities (unpublished data). The models for cariprazine, DCAR, and DDCAR also incorporated the metabolic interconversion of the 3 moieties (Figure 1).

The PBPK models for cariprazine, DCAR, and DDCAR were developed independently using parameter values obtained from preclinical in vitro and in vivo analyses [5]. Information from in vitro human liver microsome and human recombinant enzyme assessments was also incorporated (unpublished data). The three models were then simulated in a linked fashion where DCAR was formed by metabolism from cariprazine and DDCAR from DCAR in a sequential fashion.

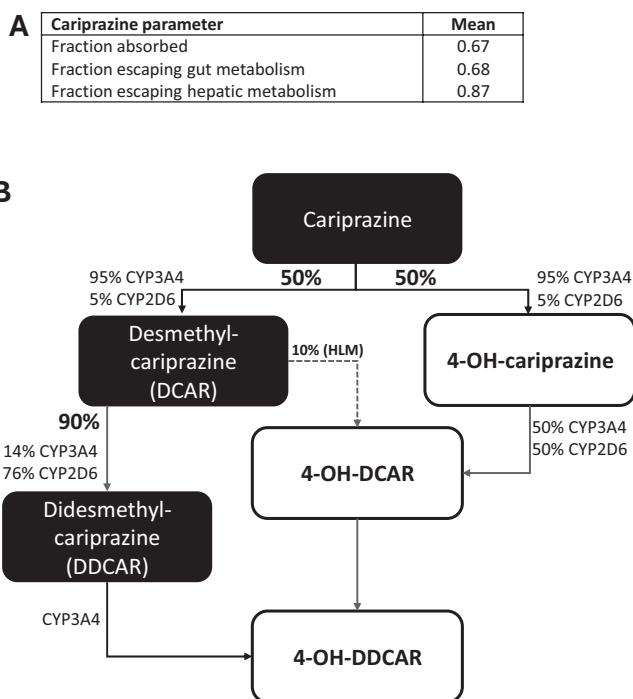


FIGURE 1 | (A) Key cariprazine absorption-related PK parameters. (B) Overview of the metabolic pathways of cariprazine. Relative contributions of cytochrome P450 (CYP) enzymes are shown and were determined from in vitro and in vivo studies (unpublished data). Solid blocks represent moieties included in the final physiologically-based pharmacokinetic (PBPK) model. Open blocks are shown for pathway completeness but were not included in PBPK modeling. 4-OH-DCAR, 4-hydroxy desmethyl-cariprazine; 4-OH-DDCAR, 4-hydroxy didesmethyl-cariprazine; HLM, human liver microsome clearance.

A first-order absorption model was utilized for model simplicity and because cariprazine has high permeability. Permeability of cariprazine was incorporated into the PBPK model using a calibrated Canine Carcinoma (Caco-2) cells permeability value of 3.5×10^{-6} cm/s, leading to an estimated human effective permeability (P_{eff}) of 0.83×10^{-4} cm/s in the jejunum I sub compartment. The resulting absorption rate (k_a) for cariprazine was similar to that obtained in a population pharmacokinetic analysis [7].

Cariprazine, DCAR, and DDCAR are highly bound to plasma proteins (91%–97%) [1]. For the current PBPK model, the cariprazine fraction unbound in the plasma ($f_{u,plasma}$) was input as 0.035. Cariprazine, DCAR, and DDCAR estimated volumes of distribution (V_d) were estimated in Simcyp using the Rogers–Rowland method [10]. V_d used in the PBPK models was 7.9, 9.8, and 3.3 L/kg for cariprazine, DCAR, and DDCAR, respectively, which were similar to those obtained in a prior population pharmacokinetic study using a combined model [7].

Cariprazine, DCAR, and DDCAR are not substrates of any drug transporters [1]. A human ADME study showed 3.7% and 1.2% elimination of unchanged cariprazine in feces and urine, respectively (unpublished data). DCAR elimination in feces and urine was 0.7% and 0.4%, respectively, and not included in the model. DDCAR biliary ($0.046 \mu\text{L}/\text{min}/10^6$ in hepatocytes) and

urinary ($0.04 \text{ L}/\text{h}$) elimination were 3.5% and 4%, respectively, and were included in the DDCAR PBPK model.

2.1.1 | Virtual Population Used in Model Simulations

All PBPK model simulations were performed using the Sim-Healthy virtual population (50% female, 20–50 years of age) in a fasted state unless specified otherwise. Clinical studies in patients were also simulated using the Sim-Healthy virtual population as PK profiles are similar in healthy participants and patients with schizophrenia [2].

2.1.2 | Model Calibration

Following initial development, the PBPK model was calibrated with existing clinical data that included cariprazine pharmacokinetic data from healthy participants and patients with schizophrenia (Table S1) [7]. Final PBPK model input parameters for the cariprazine, DCAR, and DDCAR models are provided in Tables S2 and S3.

2.1.3 | Model Verification

The PBPK models for cariprazine, DCAR, and DDCAR were verified using clinical data from four Phase 1 studies of single agent cariprazine in schizophrenia patients. Model verification with a study in Japanese patients with schizophrenia was performed using the Simcyp Japanese virtual population [8]. The ability of the PBPK model to predict clinically observed pharmacokinetic data was based on the % prediction error (%PE; Equation 1) for the relevant pharmacokinetic parameters (e.g., C_{max} and AUC). The average %PE was computed as the mean of the absolute values of individual %PE (Abs(%PE)). Based on established predictability criteria for pharmacokinetic parameters [11] and because the models were developed using data from multiple clinical studies and from both patients and healthy participants, an acceptance criterion with average Abs(%PE) $\leq 100\%$ instead of $\leq 50\%$ was justified [12, 13].

$$\%PE = \frac{(\text{Predicted mean} - \text{Observed mean})}{\text{Observed mean}} \times 100 \quad (1)$$

The calibrated PBPK models were also verified for their ability to accurately predict DDIs with CYP3A4 inhibitors. Model-predicted cariprazine exposures in the presence and absence of ketoconazole were obtained using the validated Simcyp default perpetrator model for ketoconazole. Predictions were compared to clinically observed cariprazine exposures in the presence and absence of ketoconazole based on the study design described in Figure 2A [1, 14]. Cariprazine exposures in the presence of erythromycin were also predicted using the Symcyp default model with adjustment to the erythromycin fraction absorbed as needed to best match clinical erythromycin exposures in an erythromycin-cariprazine DDI study (Figure 2B) [3]. For AUC and C_{max} ratios, the variable prediction-margin was examined and the model deemed acceptable if predicted ratios were within the limits of success (Equation 2).

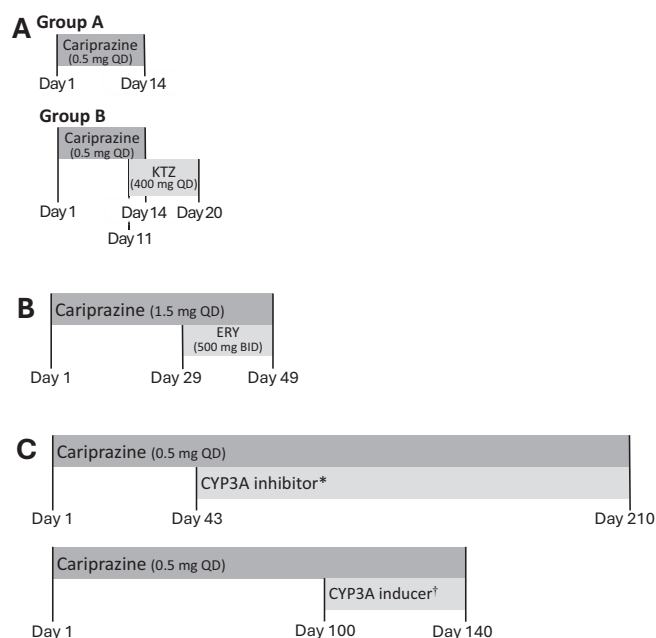


FIGURE 2 | (A) Clinical drug–drug interaction study design with KTZ, a strong CYP3A inhibitor (parallel group design) [14]. Group A received 0.5 mg QD cariprazine on Days 1–14; Group B received 0.5 mg cariprazine QD on Days 1–14 and 400 mg QD KTZ on Days 11–20. (B) Clinical drug–drug interaction study with ERY, a moderate CYP3A inhibitor (single arm, fixed sequence design). All participants received 1.5 mg QD cariprazine on Days 1–49 and 500 mg BID ERY on Days 29–49 [1, 3]. (C) Cariprazine drug–drug interaction simulation designs for strong, moderate, and weak CYP3A inhibitors and CYP3A inducers. *KTZ (400 mg QD), fluconazole (200 mg QD), cimetidine (400 mg Q2D [800 mg/day]). †rifampin (600 mg QD), efavirenz (600 mg QD), phenytoin (300 mg QD). ERY, erythromycin; KTZ, ketoconazole; QD, once per day; Q2D, once every 2 days.

$$\text{Limit} = \frac{1 + 2(R_{\text{obs}} - 1)}{R_{\text{obs}}} \quad (2)$$

where R_{obs} is the observed DDI ratio (> 1 for inhibition; for induction equals $1/[\text{observed DDI ratio}]$). The upper and lower limits of acceptance were then calculated as $R_{\text{obs}} \times \text{Limit}$ and $R_{\text{obs}}/\text{Limit}$, respectively [15].

2.2 | Model Simulations for New Drug–Drug Interaction Scenarios

To simulate the interaction effects on cariprazine, strong, moderate, and weak CYP3A inhibitors/inducers were selected using US FDA recommendations for DDI studies [16]. Model simulations for DDI predictions and dosing recommendations were performed using validated Simcyp default perpetrator models for ketoconazole, fluconazole, cimetidine, rifampin, efavirenz, and phenytoin. Figure 2C summarizes DDI simulation designs to model cariprazine exposures with coadministration of CYP3A inhibitors (ketoconazole, fluconazole, and cimetidine) and CYP3A inducers (rifampin, efavirenz, and phenytoin). Total cariprazine exposure (sum of cariprazine, DCAR, and DDCAR exposure) was used as a single PK parameter to represent the total pharmacological activities of three active species and to

measure the net effects of CYP3A4 inhibitors or inducers on them. Follow-up sensitivity analyses were performed on parameters that required adjusting based on clinical data.

3 | Results

3.1 | Physiologically-Based Pharmacokinetic Model

Cariprazine, DCAR, and DDCAR models were developed using the combined bottom-up and top-down approach. Final calibrated model input parameters are summarized in Tables S2 and S3. The model adequately recovered observed concentration-time profiles for cariprazine, DCAR, and DDCAR based on visual inspections (Figure 3, Figure S1) and pharmacokinetic parameters for cariprazine, DCAR, DDCAR, and total cariprazine (cariprazine + DCAR + DDCAR; Table S4) across clinical trials. Model verification on separate data sets confirmed that all models adequately predicted in vivo observations (Tables S5–S8). The ability of the model to predict DDIs was also illustrated by comparing model predictions to in vivo data of cariprazine and erythromycin coadministration and cariprazine and ketoconazole coadministration. In vitro estimates of f_m did not adequately capture DDI ratios for DCAR in the presence of ketoconazole. Therefore, f_m for CYP3A4 and CYP2D6 in the formation of DDCAR (from DCAR) was optimized. Adjustment was also needed in the erythromycin fraction absorbed to better match the lower-than-expected observed erythromycin exposures [3]. Model predictions all passed acceptance criteria (Table 1).

3.1.1 | Sensitivity Analyses

Sensitivity analyses were conducted to assess PBPK model parameter impact on DDI model predictions, which are the key model application results. Cariprazine permeability was measured at 14.2×10^{-6} cm/s (Caco-2 cell assay) but was calibrated to 3.5×10^{-6} cm/s to capture in vivo cariprazine fraction absorbed (f_a) and observed C_{max} . Sensitivity analysis results for cariprazine permeability are summarized in Table S9. The estimated f_a based on mass balance data in humans was approximately 0.9 and the measured C_{max} after a single 1.5 mg dose of CAR was 1.64 ng/mL. The CYP3A4 and CYP2D6 percent contributions to cariprazine and DCAR metabolism were also calibrated during model development. Sensitivity analyses supported recovery of observed ketoconazole DDI ratios using the CYP3A4 and CYP2D6 metabolic contributions of the final model (CYP3A4: cariprazine→DCAR: 95%, DCAR→DDCAR: 14%; CYP2D6: cariprazine→DCAR: 5%, DCAR→DDCAR: 76%; Table S10).

3.2 | Model Simulation of New DDI Scenarios

The effects of strong, moderate, and weak CYP3A4 inhibitors on cariprazine and metabolite exposures over 6 months of coadministration were simulated using appropriate compounds (based on US FDA perpetrator index [16]). The effects of ketoconazole and fluconazole on total cariprazine exposure were time-dependent (Table S11). At the end of the 6-month coadministration, ketoconazole, fluconazole, and cimetidine (weak CYP3A4 inhibitor)

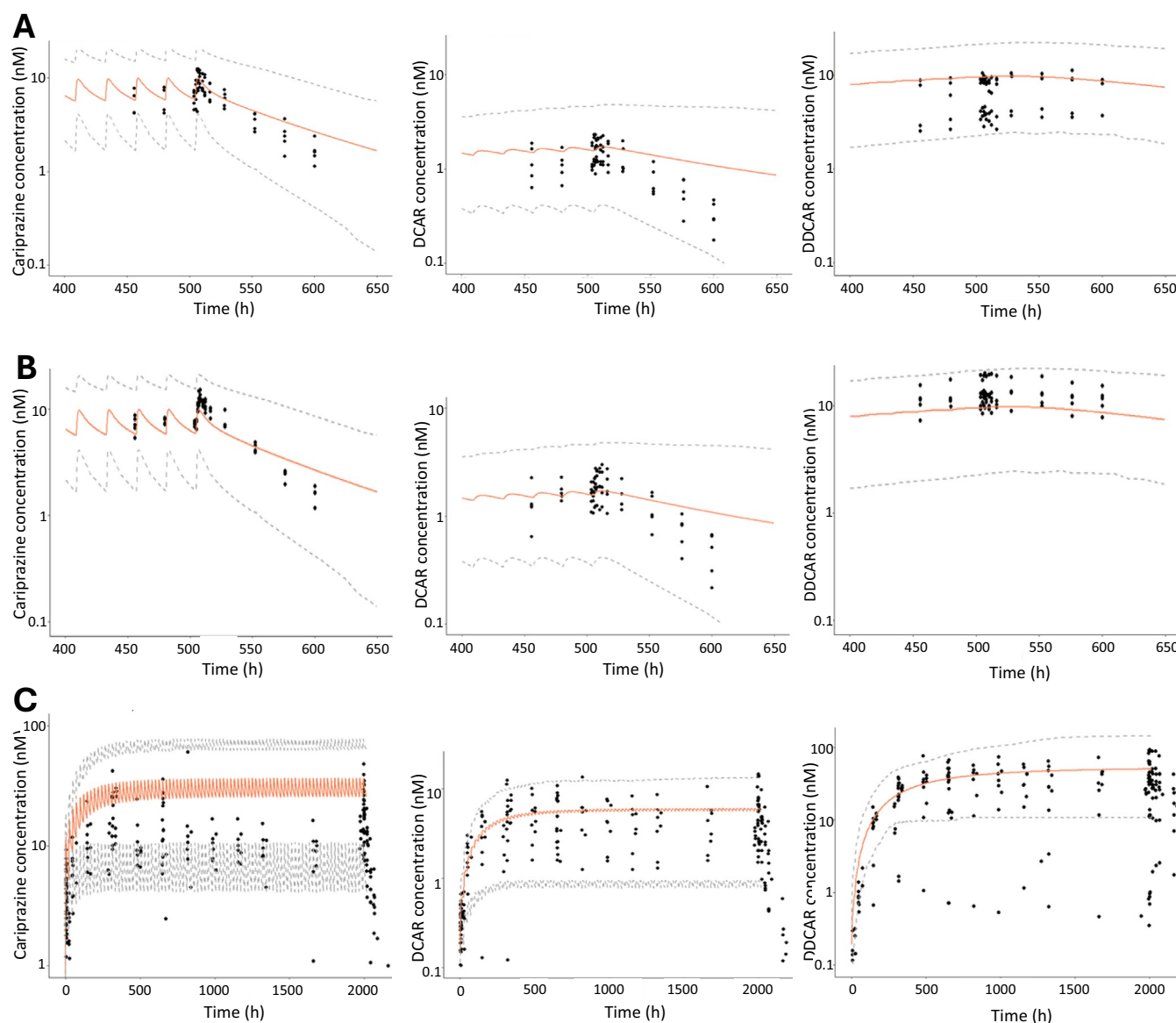


FIGURE 3 | Model verification: Representative predicted vs. observed concentration-time profiles after repeated dosing of cariprazine. (A) Study 8, Cohort A: 0.5mg QD Days 1–4 → 1.0mg QD Days 5–8 → 1.5mg QD Days 9–22. (B) Study 8, Cohort B: 1.0mg QD Days 1–4 → 1.5mg QD Days 5–8 → 2.0mg Days 9–22. (C) Study 9: 1.5mg QD Day 1 → 3.0mg QD Days 2–84. Solid orange lines represent model-predicted concentration; dashed lines represent predicted 95th percentile; circles represent clinically observed mean concentration. CAR, cariprazine; DCAR, desmethyl-cariprazine; DDCAR, didesmethyl-cariprazine; QD, once per day.

increased the $C_{\max}$ and AUC_{τ} of total cariprazine by up to 6.0-fold, 2.9-fold, and 1.1-fold, respectively (Table 2). The exposures of cariprazine, DCAR, and DDCAR were increased to a different extent by these CYP3A4 inhibitors. Conversely, all CYP3A4 inducers decreased total cariprazine $C_{\max}$ and AUC_{τ} ; rifampin by approximately 81%, efavirenz by 66%, and phenytoin by 61%.

3.3 | Cariprazine Dose Adjustments

3.3.1 | Patients on a Stable Cariprazine Dose With Initiation of a CYP3A4 Inhibitor

Dose adjustments were simulated to guide dosing recommendations in patients initiating a moderate or strong CYP3A4 inhibitor while on a stable dose of cariprazine. Based on the

estimated effect of CYP3A4 inhibitors on cariprazine exposure and lowest available commercial dose strength of 1.5 mg (Table 2), a 3- to 8-fold and 2- to 4-fold cariprazine dose reduction was simulated for strong (ketoconazole) and moderate (fluconazole) CYP3A4 inhibitor coadministration, respectively. Available dose strengths and approved dose ranges (schizophrenia and bipolar mania: 1.5–6 mg QD, bipolar depression and MDD: 1.5–3 mg QD) guided investigated cariprazine dose adjustment strategies. Of the dose adjustment strategies modeled, a 6-fold and 3-fold decrease in cariprazine dosing led to exposure ratios of approximately one in the presence of ketoconazole and fluconazole, respectively, compared to when cariprazine was administered alone (Table S11; Figure 4). A 3-fold decrease in cariprazine dose for 1.5 mg QD and an 8-fold decrease for 6 mg QD when initiating ketoconazole still rendered the total cariprazine exposure within the safe and

TABLE 1 | Model verification: Predicted and observed impact of CYP3A4 inhibitors on cariprazine and active metabolite exposures.

Species	Pharmacokinetic parameter	Observed ^a	Model predictions ^a	Acceptance criteria ^b
Ketoconazole (strong CYP3A4 inhibitor)				
Cariprazine	$C_{\max}$ ratio	3.42	2.41	2.0–5.84
	AUC_{τ} ratio	3.88	2.84	2.23–6.76
DCAR	$C_{\max}$ ratio	0.65	0.65	0.48–0.88
	AUC_{τ} ratio	0.68	0.66	0.52–0.90
DDCAR	$C_{\max}$ ratio	1.43	1.15	1.10–1.86
	AUC_{τ} ratio	1.43	1.14	1.10–1.86
Total cariprazine	$C_{\max}$ ratio	2.24	1.73	1.44–3.48
	AUC_{τ} ratio	2.14	1.78	1.40–3.28
Erythromycin (moderate CYP3A4 inhibitor)				
Cariprazine	$C_{\max}$ ratio	1.34	1.42	1.07–1.68
	AUC_{τ} ratio	1.38	1.50	1.08–1.76
DCAR	$C_{\max}$ ratio	1.16	1.06	1.02–1.32
	AUC_{τ} ratio	1.11	1.07	1.01–1.22
DDCAR	$C_{\max}$ ratio	1.62	1.32	1.17–2.24
	AUC_{τ} ratio	1.60	1.32	1.16–2.20
Total cariprazine	$C_{\max}$ ratio	1.48	1.35	1.12–1.96
	AUC_{τ} ratio	1.49	1.37	1.12–1.98

Note: DCAR, desmethyl-cariprazine; DDCAR, didesmethyl-cariprazine; total cariprazine, sum of cariprazine, DCAR, and DDCAR.

^aGeometric mean ratio.

^bBased on methods of Guest et al. 2011.

TABLE 2 | Predicted effect of CYP3A4 inhibitor/inducer coadministration on cariprazine and active metabolite exposure.

DDI scenario	Parameter	Cariprazine	DCAR	DDCAR	Total cariprazine
CYP3A4 inhibitor coadministered with cariprazine ^a					
Ketoconazole (400 mg QD)	$C_{\max}$ ratio	5.79	1.47	4.93	5.54
	AUC_{τ} ratio	7.35	1.49	4.92	5.96
Fluconazole (200 mg QD)	$C_{\max}$ ratio	2.84	1.21	2.71	2.73
	AUC_{τ} ratio	3.39	1.25	2.72	2.88
Cimetidine (400 mg BID)	$C_{\max}$ ratio	1.23	1.32	1.05	1.13
	AUC_{τ} ratio	1.22	1.34	1.05	1.12
CYP3A4 inducer coadministered with cariprazine ^b					
Rifampin (600 mg QD)	$C_{\max}$ ratio	0.15	0.66	0.21	0.20
	AUC_{τ} ratio	0.08	0.61	0.19	0.19
Efavirenz (600 mg QD)	$C_{\max}$ ratio	0.38	0.78	0.33	0.37
	AUC_{τ} ratio	0.27	0.75	0.32	0.34
Phenytoin (300 mg QD)	$C_{\max}$ ratio	0.27	0.83	0.44	0.39
	AUC_{τ} ratio	0.21	0.79	0.44	0.39

Note: All ratios are reported as geometric mean ratios.

Abbreviations: AUC_{τ} , area under the concentration-time curve during a dosing interval; BID, twice a day; $C_{\max}$, maximum plasma concentration of cariprazine; QD, once per day.

^aSimulations performed with cariprazine dosing of 0.5 mg QD for 210 days; CYP3A4 inhibitor initiated on Day 43.

^bSimulations performed with cariprazine dosing of 0.5 mg QD for 140 days; CYP3A4 inducer initiated on Day 100.

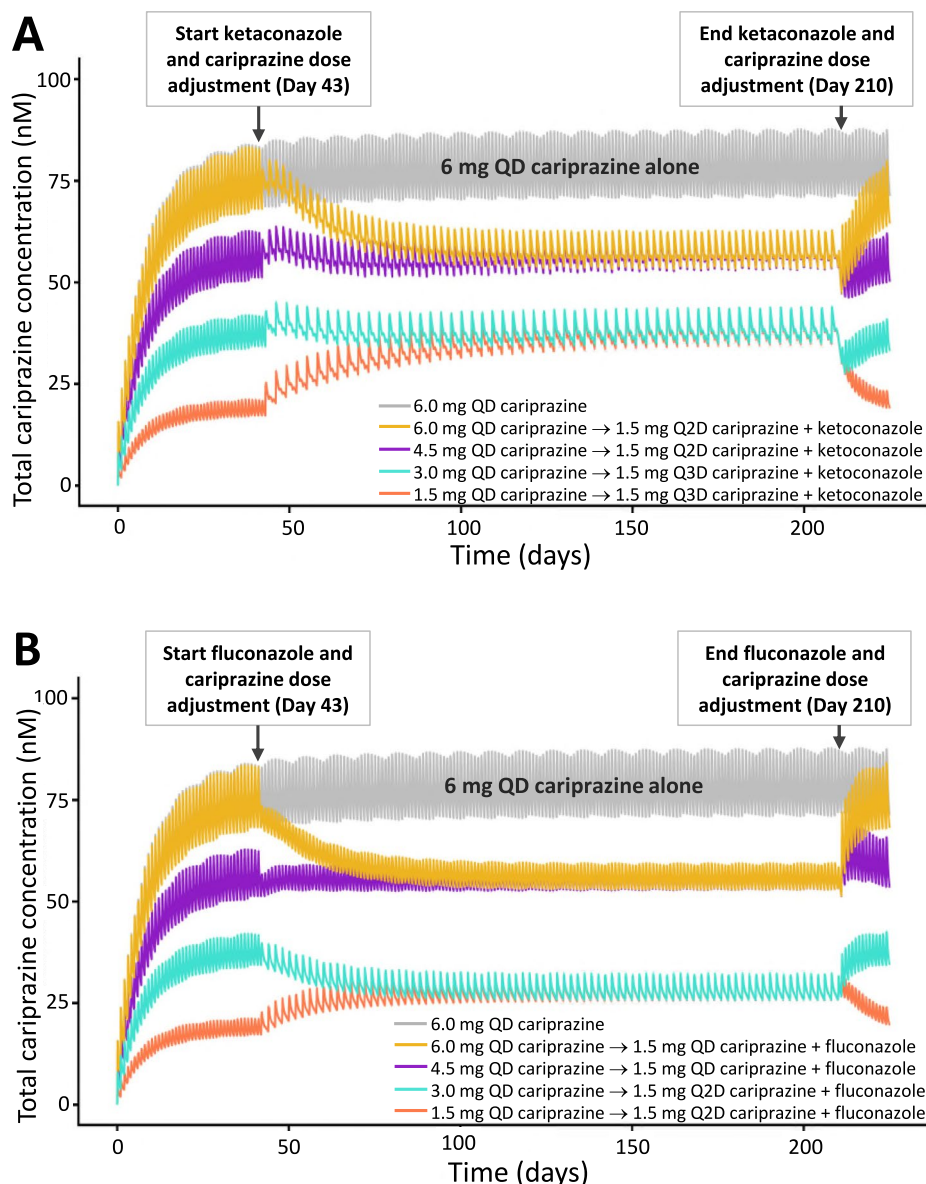


FIGURE 4 | Physiologically-based pharmacokinetic model simulations of total cariprazine before and after initiation of a strong (ketoconazole, 400 mg QD; A) and moderate (fluconazole, 200 mg QD; B) CYP3A4 inhibitor with cariprazine dose adjustment. As indicated by the arrows, CYP3A4 coadministration and cariprazine dose reduction occurred on Days 43–210. The safe and efficacious exposure range for total cariprazine is represented by the steady-state total cariprazine exposure for cariprazine dosing of 1.5–6 mg QD before the initiation of CYP3A4 inhibitor administration. Simulations support a 3-fold and 6-fold cariprazine target dose reduction when concomitantly administered with a moderate (B) and strong (A) CYP3A4 inhibitor. QD, once daily; Q2D, once every 2 days; Q3D, once every 3 days.

effective exposure range of the recommended dose range. Similarly, a 2-fold dose reduction for 1.5 mg QD and 4-fold dose reduction for 3 or 6 mg QD when initiating fluconazole rendered the total cariprazine exposure within the safe and effective exposure range of the recommended dose range.

3.3.2 | Patients on a Stable CYP3A4 Inhibitor Dose With Initiation of Cariprazine

A cariprazine starting dose of 1.5 mg QD is recommended for all patients, regardless of indication. The daily dose can be titrated up to 3 mg QD for bipolar depression and MDD and up to 6 mg QD for schizophrenia and bipolar mania, dependent upon

clinical response and cariprazine tolerability [1]. Therefore, simulations were based on a stable cariprazine dose of 1.5 mg QD (Figure 5). As the PBPK model predicted, patients on a stable dose of ketoconazole (400 mg QD) who initiated cariprazine with a 3-fold dose reduction via an increase in the dosing interval (1.5 mg once every 3 days [Q3D]) had similar total cariprazine (cariprazine + DCAR + DDCAR) exposure as patients on cariprazine only (1.5 mg QD) over the first 2 weeks of cariprazine administration (Figure 5A). Although the predicted steady-state total cariprazine exposure remained higher with this dose adjustment, exposures were well within those in the approved dose range (1.5–6 mg QD depending on indication) [1]. Findings were similar for patients on a stable dose of fluconazole (200 mg QD) who initiated a 2-fold dose reduction via an

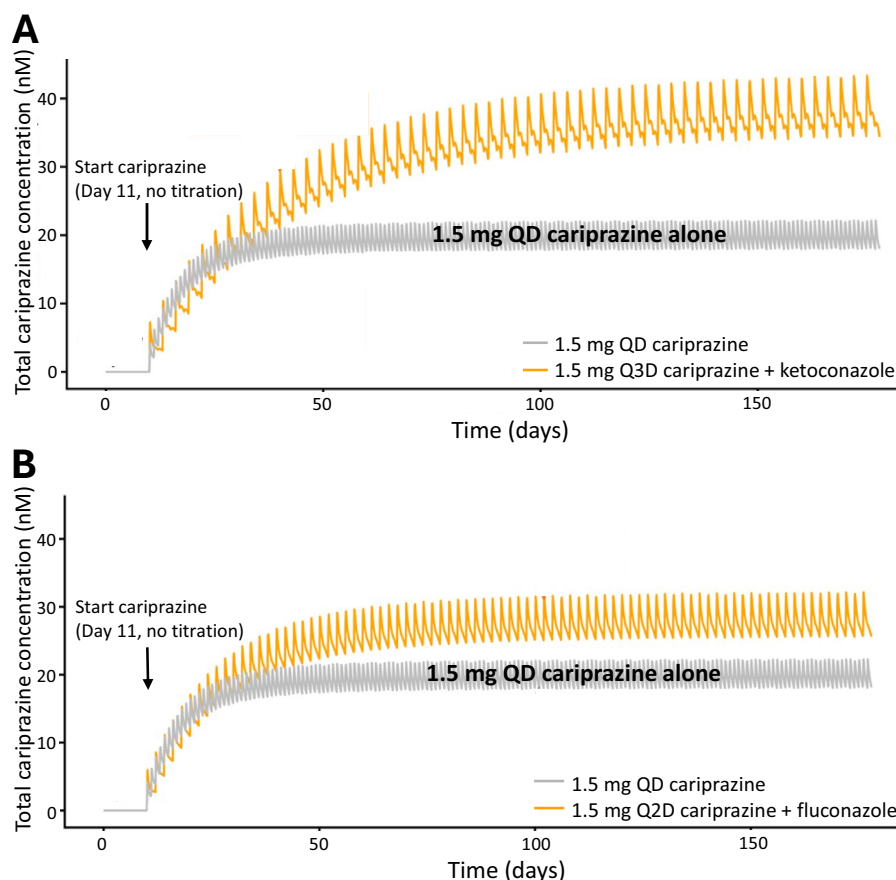


FIGURE 5 | Physiologically-based pharmacokinetic model simulations of total cariprazine in patients initiating cariprazine with or without a concomitant CYP3A4 inhibitor. (A) Initiating cariprazine at 1.5 mg Q3D on Day 11 on a stable 400 mg ketoconazole QD dose (started from Day 1) compared to initiating cariprazine at 1.5 mg QD in the absence of ketoconazole. (B) Initiating cariprazine at 1.5 mg Q2D on Day 11 on a stable 200 mg fluconazole QD dose (started from Day 1) compared to initiating cariprazine at 1.5 mg QD alone. QD, once per day; Q2D, every 2 days; Q3D, every 3 days.

increase in the dosing interval (1.5 mg once every 2 days [Q2D]; Figure 5B).

4 | Discussion

The main objective of the current analysis was to guide cariprazine dose adjustment and prescribing recommendations using predicted steady-state plasma concentrations under the scenario of prolonged cariprazine and CYP3A4 inhibitor coadministration. This information was needed to determine optimal dosing adjustments due to the long half-life of total cariprazine and the common longer-term use of CYP3A4 inhibitors [9]. In particular, the antifungal agent, ketoconazole, is a strong CYP3A4 inhibitor and has a usual treatment duration of up to 6 months [9]. Dedicated DDI studies between cariprazine and ketoconazole (unpublished data) and between cariprazine and erythromycin [3] have been completed, but the durations for co-administration with cariprazine were relatively short (4 days and 21 days, respectively) compared to the long half-life (1 to 3 weeks) of DDCAR, a main active metabolite of cariprazine. Under the CYP3A4 inhibition scenario, the DDCAR half-life is expected to be significantly longer. Due to the impracticable challenge of conducting a clinical cariprazine-CYP3A4 inhibitor DDI study over several months to achieve steady-state conditions, we developed PBPK

models for cariprazine and its two active metabolites (DCAR and DDCAR) based on available clinical data. These models served as the basis for PBPK model simulations to evaluate the impact of untested potential DDIs with cariprazine.

Individual and sequentially linked PBPK models for cariprazine and its active metabolites (DCAR and DDCAR) were developed and adequately characterized the pharmacokinetics of all three moieties across multiple Phase 1 and 2 studies. Despite large interstudy variation, the final model adequately predicted cariprazine, DCAR, and DDCAR exposures for single- and multiple-dose clinical studies in healthy participants and patients with schizophrenia with an average parameter %PE $\leq 100\%$ or within 2-fold. More importantly, all model DDI predictions of exposure ratios in the presence and absence of moderate and strong CYP3A4 inhibitors were within acceptance criteria.

In this PBPK analysis, total cariprazine (sum of cariprazine, DCAR, and DDCAR) exposure in plasma was selected to correlate to the sum of pharmacological activities of three active species. Although theoretically, the sum of unbound potency-corrected active species may better correlate with pharmacological response of a drug, whether it is better or even feasible in the case of cariprazine is questionable for the following reasons. First, the exact mechanism of action of cariprazine is not

entirely known. Cariprazine efficacy could be mediated through a combination of partial agonist activity at central dopamine D_2 and D_3 receptors and serotonin 5-HT_{1A} receptors and antagonist activity at serotonin 5-HT_{2A} receptors [1]. Cariprazine and its two active metabolites are known to have similar but numerically different binding affinity to dopamine D_2 and D_3 receptors [5]. Therefore, it is difficult to identify a single potency value for each active species. Second, DDCAR is thought to be the predominant active species that drives efficacy and safety because DDCAR is the most abundant circulating active species at steady state (based on mean trough concentrations: ~23% cariprazine, ~7% DCAR, and ~70% DDCAR [7]). Therefore, using the sum of cariprazine, DCAR, and DDCAR concentrations is a simple and reasonable exposure parameter to predict efficacy and safety. Correcting the minor differences in protein binding and potency (if possible) would have minor impact on interpreting the results.

Initial product labeling in 2015 recommended that cariprazine dose be cut in half with initiation of a strong CYP3A4 inhibitor and did not include specific recommendations for moderate CYP3A4 inhibitors [6]. The recommendation was based on findings from a limited clinical DDI study, where coadministration of ketoconazole with cariprazine for 4 days increased total cariprazine C_{max} and AUC_{0-24h} by approximately 100% [1]. Given the long half-life of DDCAR (1–3 weeks), it was unclear what effect a strong CYP3A4 inhibitor would have on total cariprazine during a coadministration over months. Practically, it is very challenging to conduct a clinical DDI study that requires study participants to take cariprazine with a strong CYP3A4 inhibitor for months. Results from the current analyses, which simulated the effects of CYP3A4 inhibitors on total cariprazine exposure over 6 months to ensure a new steady state was reached, provided a full picture on the DDI, and allowed for an optimized dose adjustment strategy for patients coadministered cariprazine with a strong or moderate CYP3A4 inhibitor. As a result of current study findings, cariprazine dose adjustment recommendations needed to be updated, ideally recommending a 6-fold decrease with strong CYP3A4 coadministration and a 3-fold decrease in dosage with moderate CYP3A4 coadministration. In consideration of the lowest available commercial cariprazine dose (1.5 mg capsule) and a convenient dosing schedule for patients on a stable cariprazine dose prior to CYP3A4 inhibitor initiation, a 3- to 8-fold cariprazine dose reduction is recommended with concomitant use of a strong CYP3A4 inhibitor; a 2- to 4-fold reduction with concomitant use of a moderate CYP3A4 inhibitor [1]. With the recommended cariprazine dose adjustment, total cariprazine exposure in the presence of a strong or moderate CYP3A4 inhibitor is expected to be maintained in the safe and effective range established from the approved dose range for each adult indication [1]. Although the original cariprazine dose adjustment recommendation may not lead to clinically meaningful deviation in total cariprazine exposure over several days of concomitant use with strong and moderate CYP3A4 inhibitors, greater cariprazine dose reduction recommendation based on the PBPK modeling better maintains the total cariprazine exposure during coadministration with CYP3A4 inhibitors, especially during prolonged concomitant use. This work presented here highlighted the importance of using PBPK modeling to extend the findings from a limited clinical study to understand the full extent of DDI.

According to a model credibility assessment framework published previously [17], the PBPK models of cariprazine, DCAR, and DDCAR are well verified and validated. The application of the current PBPK models to optimize dose adjustments for cariprazine when concomitantly used with strong or moderate CYP3A4 inhibitors was categorized as medium risk according to the model credibility assessment framework [17]. In terms of model influence, some clinical data on the effects of strong and moderate CYP3A4 inhibitors on cariprazine PK are available, and the PBPK models provide additional supportive evidence. On the decision consequence, current PBPK model simulations recommended greater cariprazine dose reduction in the presence of strong or moderate CYP3A4 inhibitors than initially recommended and are expected to improve product safety and maintain product efficacy when cariprazine is coadministered with a strong or moderate CYP3A4 inhibitor.

Current PBPK modeling analysis predicted that coadministration of cariprazine with rifampin (a strong CYP3A4 inducer) and efavirenz (a moderate CYP3A4 inducer) would reduce total cariprazine exposure by up to 81% and 66%, respectively. The predictions support the initial cariprazine drug label, which did not recommend the concomitant use of cariprazine and a CYP3A4 inducer. In the real world, the probability of a patient needing to take cariprazine with a strong or moderate CYP3A4 inducer is much lower compared to a CYP3A4 inhibitor. Therefore, a cariprazine dose adjustment recommendation for concomitant use of CYP3A4 inducers was not sought.

The current analysis had limitations. First, existing DDI clinical data that informed the model were from shorter dosing duration studies than the simulations conducted using the current model for both moderate (21 days [3] vs. 168 days) and strong (4 days [unpublished data] vs. 168 days) CYP3A4 inhibitors. Therefore, cariprazine exposures predicted under the simulation time frame in this modeling analysis were extrapolated and may have higher uncertainty. Second, the Abs(%PE) acceptance criterion ($\leq 100\%$) of modeled PK parameters was relatively high due to source clinical data being from multiple studies. Differences between studies in cariprazine dosing, treatment duration, and study population likely lead to higher data variability. Third, CYP2D6 phenotype was not incorporated into the cariprazine PBPK model. However, this is not expected to meaningfully affect the predicted effects of CYP3A4 inhibitors or inducers on cariprazine because CYP2D6 only has a minor role in the overall metabolism of cariprazine. Prior population PK analysis has shown that CYP2D6 metabolizer status was not significantly associated with exposure levels for cariprazine, DCAR, or DDCAR [7]. In addition, CYP2D6 phenotypes did not meaningfully influence the effects of erythromycin on the PK of cariprazine, DCAR, or DDCAR [3]. Lastly, because the lowest commercially available dose strength of cariprazine is 1.5 mg, dose adjustment strategies for CYP3A4 inhibitor coadministration now recommend extending dosing interval to Q2D or Q3D with the same cariprazine dose. However, daily dosing is a simpler treatment regimen, often preferred by patients, and may lead to better medication compliance.

In conclusion, PBPK models for cariprazine, DCAR, and DDCAR were developed and validated. These models predicted DDIs between cariprazine and CYP3A4 inhibitors during both

short-term and long-term coadministration. These predictions informed dosage adjustment recommendations for patients to maintain acceptable drug exposure in the presence of strong or moderate CYP3A inhibitors and led to a US prescribing information update for cariprazine in November 2024 [1]. This is of particular importance for cariprazine patients who suffer from a variety of serious psychiatric conditions and must maintain treatment efficacy without increasing the risk of drug-related adverse effects.

Author Contributions

Wrote manuscript: H.X., M.M.H.R., P.M., and M.S. Designed research: H.X., M.M.H.R., P.M., and M.S. Performed research: H.X., M.M.H.R., and P.M. Analyzed data: M.M.H.R.

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Conflicts of Interest

All authors are employees of AbbVie Inc. and may hold stock or options in the company.

Data Availability Statement

AbbVie is committed to responsible data sharing regarding the clinical trials we sponsor. This includes access to anonymized, individual, and trial-level data (analysis data sets), as well as other information (e.g., protocols, clinical study reports, or analysis plans), as long as the trials are not part of an ongoing or planned regulatory submission. This includes requests for clinical trial data for unlicensed products and indications. These clinical trial data can be requested by any qualified researchers who engage in rigorous, independent, scientific research, and will be provided following review and approval of a research proposal, Statistical Analysis Plan (SAP), and execution of a Data Sharing Agreement (DSA). Data requests can be submitted at any time after approval in the US and Europe and after acceptance of this manuscript for publication. The data will be accessible for 12 months, with possible extensions considered. For more information on the process or to submit a request, visit the following link: <https://www.abbvieclinicaltrials.com/hcp/data-sharing/html>.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Appendix S1:** psp470236-sup-0001-AppendixS1.docx.